

Hadoop Word Count Using MapReduce

1. Objective

To create a text file, store it in HDFS, execute the Word Count MapReduce program using Hadoop, and view the word frequency results.

2. Procedure

Step 1: Create a New Text File

```
touch prac.txt
```

Data Science is the process of collecting, analyzing, and interpreting data to discover useful insights and support decision-making. It combines statistics, programming, machine learning, and domain knowledge to solve real-world problems. Data Science is widely used in industries such as healthcare, finance, marketing, and technology to improve efficiency and predict future trends.

Step 2: Verify the File Contents

```
cat prac.txt
```

Step 3: Upload the File to HDFS

```
hdfs dfs -put prac.txt /demo
```

Step 4: Execute the Word Count Program

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar  
wordcount /demo /output
```

Step 5: Check the Output Directory

```
hdfs dfs -ls /output
```

Step 6: View the Word Count Results

```
hdfs dfs -cat /output/part-r-00000
```

3. Source Code

Input File (prac.txt)

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Commands Used

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```

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```

```
hdfs dfs -ls /output
```

```
hdfs dfs -cat /output/part-r-00000
```

4. Screenshots

Creating the Input File

```
pratham@LAPTOP-PL4G895R:~$ touch prac.txt  
pratham@LAPTOP-PL4G895R:~$ nano prac.txt  
pratham@LAPTOP-PL4G895R:~$ cat prac.txt  
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```

Verifying the File

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pratham@LAPTOP-PL4G895R:~$ touch prac.txt
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```

Running Word Count

```
pratham@LAPTOP-PL4G895R:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /demo /output
2026-06-21 12:39:21,715 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2026-06-21 12:39:22,457 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-06-21 12:39:22,571 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-06-21 12:39:22,571 INFO impl.MetricsSystemImpl: JobTracker metrics system started
org.apache.hadoop.mapred.FileAlreadyExistsException: Output directory hdfs://localhost:9000/output already exists
    at org.apache.hadoop.mapreduce.lib.output.FileOutputFormat.checkOutputSpecs(FileOutputFormat.java:164)
    at org.apache.hadoop.mapreduce.JobSubmitter.checkSpecs(JobSubmitter.java:277)
    at org.apache.hadoop.mapreduce.JobSubmitter.submitJobInternal(JobSubmitter.java:143)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1678)
    at org.apache.hadoop.mapreduce.Job$11.run(Job.java:1675)
    at java.base/java.security.AccessController.doPrivileged(Native Method)
    at java.base/javax.security.auth.Subject.doAs(Subject.java:423)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1899)
    at org.apache.hadoop.mapreduce.Job.submit(Job.java:1675)
    at org.apache.hadoop.mapreduce.Job.waitForCompletion(Job.java:1696)
    at org.apache.hadoop.examples.WordCount.main(WordCount.java:87)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:566)
    at org.apache.hadoop.util.ProgramDriver$ProgramDescription.invoke(ProgramDriver.java:71)
    at org.apache.hadoop.util.ProgramDriver.run(ProgramDriver.java:144)
    at org.apache.hadoop.examples.ExampleDriver.main(ExampleDriver.java:74)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
    at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:62)
    at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
    at java.base/java.lang.reflect.Method.invoke(Method.java:566)
    at org.apache.hadoop.util.RunJar.run(RunJar.java:328)
    at org.apache.hadoop.util.RunJar.main(RunJar.java:241)
```

Screenshot 5: Checking Output Directory

```
pratham@LAPTOP-PL4G895R:~$ hdfs dfs -ls /output
2026-06-21 12:40:12,430 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 1 pratham supergroup 0 2026-06-11 04:31 /output/_SUCCESS
-rw-r--r-- 1 pratham supergroup 25 2026-06-11 04:31 /output/part-r-00000
```

Screenshot 6: Viewing Word Count Results

```
pratham@LAPTOP-PL4G895R:~$ hdfs dfs -cat /output/part-r-00000
2026-06-21 12:41:42,738 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hadoop 1
hello 2
spark 1
```

5. Conclusion

The Hadoop MapReduce Word Count application was successfully executed. The input text file was uploaded to HDFS, processed using the built-in Hadoop Word Count program, and the output containing the frequency of each word was generated and stored in the HDFS output directory. This experiment demonstrates the basic working of Hadoop MapReduce for distributed data processing.